MILITARY SPECIFICATION SHEET

METER, TIME TOTALIZING, MINIATURE, DIGITAL, 5 VDC, PRINTED CIRCUIT MOUNT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirement for acquiring the meter described herein shall consist of this specification and the latest issue in effect of MIL-M-7793.

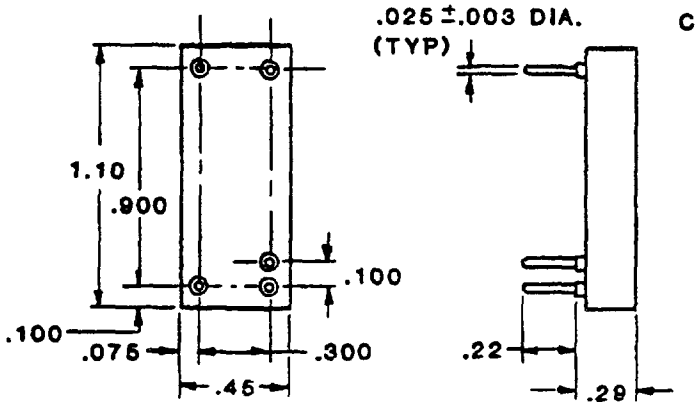


FIGURE 1. Bottom view.

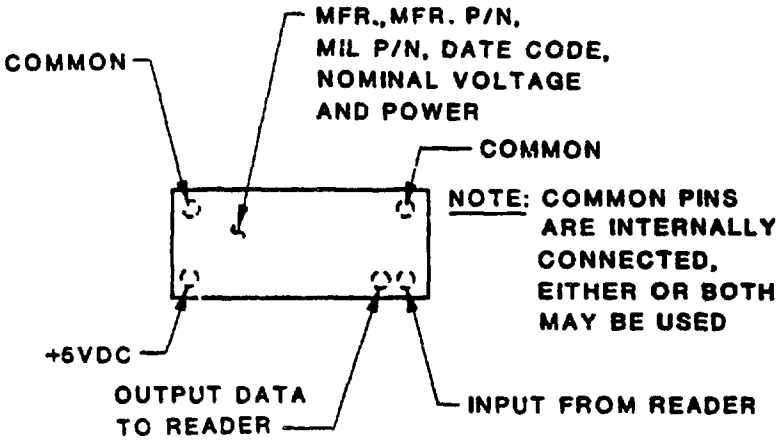


FIGURE 2. Top view and pin assignment.

AMSC N/A FSC 6645

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:00Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions are in inches. Unless otherwise specified, tolerances are ±.02 for two-place decimals and ±.015 for three-place decimals.

Performance:

Dielectric strength: The meter shall be subjected to 600 VRMS (room) and 350 VRMS (altitude). The test voltage shall be applied between the power terminals (+5 VDC and common) and an external ground which contacts the meter case on the five sides without terminals.

Life accuracy: ±0.1% from -65° to +125°C and 4.5 to 10 VDC.

Power consumption: Power consumption at 5 VDC shall not exceed 2 milliwatts.

Ripple voltage: Meter shall operate normally when subjected to a 2 volt peak (4 volts peak-to-peak) ripple between 10 Hz and 10 KHz superimposed on 7.0 VDC.

Operating temperature range: -65° to +125°C.

Part number: M7793/13-1.

Time range: Total readout of the meters shall be 99999.99 hours.

Meter output: The output from the meter shall be in a serial binary coded decimal format. The output impedance shall be 100K ± 10%. A logical "zero" shall be between 0.0 and +0.2 volts. A logical "one" shall be between +3.3 and +6.6 volts. The output pin assignments are shown in Figure 2 and the output data format is shown in Figure 3.

Transient protection: No temporary or permanent degradation shall be produced in the meter when subjected to ±25 volt transients of 10 microsecond duration occurring at a 1 millisecond repetition rate.

Voltage range: Operating voltage shall be from 4.5 to 10 VDC.

Weight: Weight shall be not greater than 0.2 ounce.

Qualification tests: Qualification tests will be conducted in accordance with MIL-M-7793, except that the voltage, temperature and accuracy shall be as specified herein. Hermetic seal test is not applicable.

NOTES:

1. This unit is intended for use on printed circuit boards. All connections are expected to be made via printed wiring, and the output to be brought to a data collection point for system reading or to a single dedicated connector (see Note 2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:00Z
Check the source to verify that this is the current version before use.

2. Elapsed time is read from the M7793/13-1 meter by mating the printed wiring board connections with the M7793/12-1 reader's connector and operating the reader.

3. Digital readout and meter face plate per MIL-M-7793 are not applicable since the digital readout is provided by the separate M7793/12-1 reader.

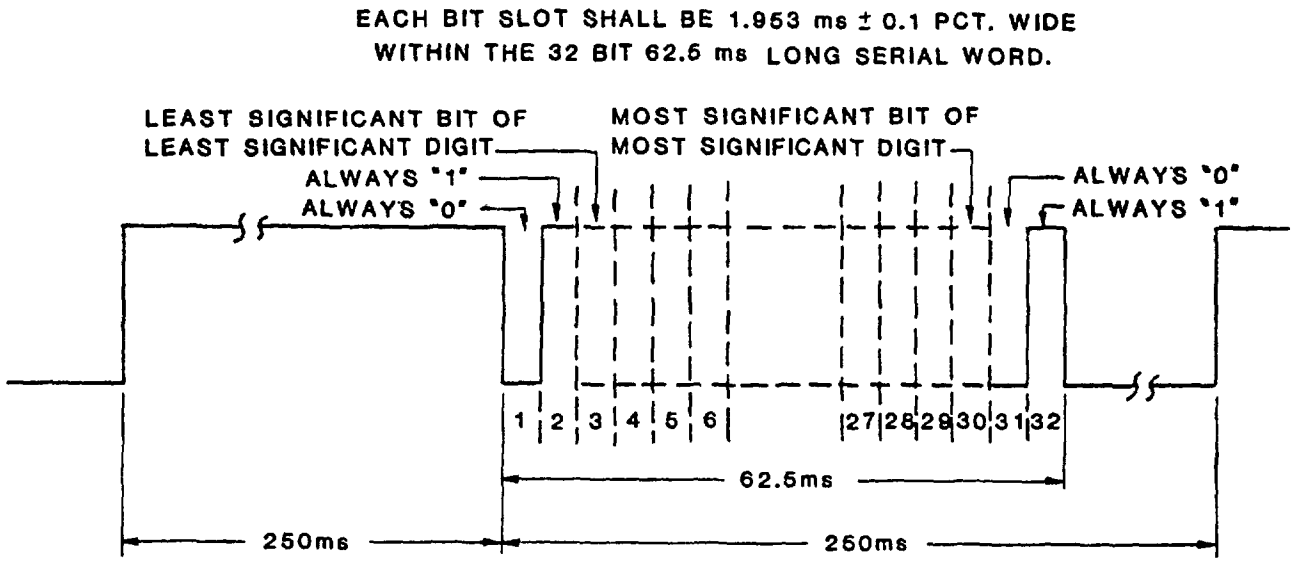


FIGURE 3. Output data format.

Custodians:
Navy - AS
Army -
Air Force - 99

Users:
Navy - SH, OS

Reviewer:
Army - MI

Preparing activity:
Navy - AS
(Project Number 6645-400)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T09:00Z
Check the source to verify that this is the current version before use.